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Review What You Learned

In this course, you explored generative AI techniques and their applications in data science. You learned about integrating generative AI into data science projects, including data augmentation and feature engineering. You also learned how to use and apply generative models for tasks such as data generation, augmentation, preparation, querying, and more.

Here are the key takeaways from this course:

Generative AI empowers data scientists to generate entirely new data, unlocking a universe of possibilities and tackling previously insurmountable challenges.

- There are four standard generative AI models: Generative adversarial networks (GANs), variational autoencoders (VAEs), autoregressive, and flow-based models.
- While GANs are great at data augmentation, VAEs are good at anomaly detection, data compression, collaborative filtering, and style transfer.
- Autoregressive models are good at text generation, speech synthesis, time series forecasting, and machine translation, and flow-based models are suitable for image and data generation and density estimation.
- Generative AI can tackle complex problems across various industries.
- Generative AI models are instrumental in tackling several data preparation and querying challenges, such as inputting missing values, detecting outliers, reducing “noise,” and translating natural language queries into equivalent SQL statements.
- Generative AI can help in exploratory data analysis or EDA using various techniques, such as statistical data description, univariate, bivariate, multivariate analysis, feature engineering, and hypothesis generation.
- Generative AI plays a significant role in developing a predictive model using various techniques, such as selecting model architecture and essential features, generating ensemble models, improving model interpretability and generalization, and preventing overfitting.
- While using generative AI models, data scientists need to look into data, models, and ethical considerations.